Light

GCSE Computer Science – Boolean Logic (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Boolean Logic

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What does a NOT gate do?

- **A.** It outputs 1 only when both inputs are 1
- **B.** It outputs 1 when at least one input is 1
- **C.** It reverses the input, changing 0 to 1 and 1 to 0
- **D.** It outputs 1 only when inputs are different

Q2 (1 mark)

When does an AND gate output 1?

- **A.** When at least one input is 1

- B. When all inputs are 1
- C. When all inputs are 0
- D. When inputs are different

Q3 (1 mark)

What is the output of an OR gate when both inputs are 0?

- A. 0
- B. 1
- C. 2
- D. It depends on the circuit

Q4 (1 mark)

An XOR gate outputs 1 when:

- A. Both inputs are 1
- B. Both inputs are 0
- C. The inputs are different (one is 0 and one is 1)
- D. At least one input is 1

Q5 (1 mark)

How many rows would a truth table have for a circuit with 2 inputs?

- A. 2 rows
- B. 8 rows
- C. 16 rows
- D. 4 rows

Q6 (1 mark)

In Boolean algebra, which symbol represents the AND operation?

- A. +
- B. .
- C. $\oplus$
- D. /

Q7 (1 mark)

In Boolean algebra, which symbol represents the OR operation?

- A. +
- B. .
- C. ×
- D. -

Q8 (1 mark)

If the input to a NOT gate is 1, what is the output?

- ○ **A.** 1
- ○ **B.** 2
- ○ **C.** It stays the same
- ○ **D.** 0

Q9 (1 mark)

What is the output of an AND gate when A = 1 and B = 0?

- ○ **A.** 1
- ○ **B.** 0
- ○ **C.** Undefined
- ○ **D.** Both 0 and 1

Q10 (1 mark)

What is the output of an XOR gate when A = 1 and B = 1?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** 2

- ○ **D.** Undefined

Q11 (1 mark)

Which symbol represents XOR in Boolean expressions?

- ○ **A.** +
- ○ **B.** .
- ○ **C.** ⊕
- ○ **D.** ∧

Q12 (1 mark)

A truth table has 8 rows. How many inputs does the circuit have?

- ○ **A.** 2 inputs
- ○ **B.** 4 inputs
- ○ **C.** 8 inputs
- ○ **D.** 3 inputs

Q13 (2 marks)

In a circuit, the output of an OR gate (with inputs A and B) feeds into an AND gate along with input C. If A = 0, B = 1, and C = 1, what is the final output?

- A. 0
- B. 1
- C. Cannot be determined
- D. 2

Q14 (2 marks)

Input A passes through a NOT gate, then the result is fed into an AND gate with input B. If A = 1 and B = 1, what is the output?

- A. 0
- B. 1
- C. Depends on the NOT gate type
- D. Cannot determine

Q15 (2 marks)

What is the value of the Boolean expression A.B when A = 1 and B = 0?

- A. 0
- B. 1
- C. A.B means A divided by B
- D. Undefined

Q16 (2 marks)

In the Boolean expression $Q = \bar{A}$, what does the overbar (line above A) represent?

- **A.** A squared
- **B.** A multiplied by itself
- **C.** The average of A
- **D.** NOT A (the inverse of A)

Q17 (2 marks)

For which input combination do OR and XOR gates give different outputs?

- **A.** A = 0, B = 0
- **B.** A = 0, B = 1
- **C.** A = 1, B = 0
- **D.** A = 1, B = 1

Q18 (2 marks)

Evaluate the Boolean expression A + B when A = 0 and B = 1.

- **A.** 0
- **B.** 1
- **C.** 2

- D. 01

Q19 (2 marks)

In an AND gate truth table, how many rows have an output of 1?

- A. 1 row
- B. 2 rows
- C. 3 rows
- D. 4 rows

Q20 (2 marks)

In an OR gate truth table (2 inputs), how many rows have an output of 1?

- A. 1 row
- B. 2 rows
- C. 3 rows
- D. 4 rows

Q21 (2 marks)

In an XOR gate truth table (2 inputs), how many rows have an output of 1?

- A. 1 row

- B. 2 rows
- C. 3 rows
- D. 4 rows

Q22 (2 marks)

Evaluate the expression $\bar{A} + B$ when A = 1 and B = 0.

- A. 0
- B. 1
- C. -1
- D. Undefined

Q23 (2 marks)

A circuit has inputs A and B. First, both inputs go into an XOR gate. The XOR output then goes into a NOT gate. If A = 0 and B = 1, what is the final output?

- A. 0
- B. 1
- C. Cannot be determined
- D. The circuit is invalid

Q24 (3 marks)

A circuit has three inputs A, B, and C. The Boolean expression is (A.B) + C. In the truth table, how many rows will have output 1?

- **A.** 4 rows
- **B.** 6 rows
- **C.** 7 rows
- **D.** 5 rows

Q25 (3 marks)

Which Boolean expression represents a circuit where A and B go into an AND gate, and the result is ORed with C?

- **A.** A + B.C
- **B.** (A.B) + C
- **C.** A.(B + C)
- **D.** (A + B).C

Q26 (3 marks)

A circuit has the expression $A \oplus \bar{B}$. When A = 1 and B = 1, what is the output?

- **A.** 0
- **B.** 1
- **C.** 2

- **D.** Cannot be evaluated

Q27 (3 marks)

A circuit has inputs A, B, and C. A and B feed into an OR gate. The OR output and C feed into an AND gate. For which combination is the final output 0?

- **A.** A = 1, B = 0, C = 1
- **B.** A = 0, B = 1, C = 1
- **C.** A = 0, B = 0, C = 1
- **D.** A = 1, B = 1, C = 1

Q28 (3 marks)

For which input values does A.B give the same result as A $\oplus$ B?

- **A.** Only when A = 0 and B = 0
- **B.** Only when A = 1 and B = 1
- **C.** When A = 0, B = 0 OR when A = 1, B = 1
- **D.** When A = 0, B = 1 OR when A = 1, B = 0

Q29 (3 marks)

A burglar alarm should sound (output 1) when the door sensor (D) is triggered OR when both the motion sensor (M) AND window sensor (W) are triggered. Which expression represents this?

- **A.** D.M.W
- **B.** D + M + W
- **C.** D + (M.W)
- **D.** (D + M).W

Q30 (3 marks)

A truth table for 3 inputs (A, B, C) shows output Q. The output Q = 1 only for these rows: (0,0,1), (0,1,1), (1,0,1), (1,1,1). Which expression matches this pattern?

- **A.** A.B.C
- **B.** A + B + C
- **C.** C
- **D.** (A.B) + C

Total: **55** marks • Time: **30 min**

↑ Back to top

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